

ASSIGNMENT #5

Course: **CHE 301. CHEMICAL ENGINEERING THERMODYNAMICS**
Term: Spring 2020
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Your name: Giselle Cuevas _____

Due Date: Saturday April 04 by 12:00 noon

Number of points.....100

Your score.....

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

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Problem 5A:

- 1) To find the fugacity of water at 250°C and 70 bar using Steam Tables we can look at the equation:

$$\int dG = \int -SdT + \int V dP$$

With constant T we get

$$\int dG = \int V dP$$

For ideal gases $V=RT/P$ so we integrate the equation and get

$$G^{IG} = C + RT \ln P \quad (eq. 1)$$

Where C is a constant of integration

For real gases we can replace pressure "P" with fugacity "f" and we get

$$G = C + RT \ln f \quad (eq. 2)$$

If we combine equation 1 and 2 we get

$$G - G^{IG} = RT \ln \frac{f}{P} \quad (eq. 3)$$

Since we can't assume that we have an ideal gas we have to modify eq.3

We know that as Pressure approaches 0, $f=P$ so we can replace equation eq.3 with

$$G_2 - G_1 = RT \ln \left(\frac{f_2}{f_1} \right) \quad (eq. 4)$$

We can now use a relatively low Pressure for our f_1 and use that pressure to calculate G_2 and we use the data given for water to solve G_2 and we can solve the fugacity (f_2) for water.

We will use the Steam tables to solve for G_1 and G_2 .

The pressures in the steam tables are in kPa so all pressures will be converted to kPa and we will convert to atm when we get the final answer.

We will convert all our temperatures to Kelvin.

The equation to solve for G is

$$G = H - ST$$

So from the steam tables we find that at 250°C and 70 bar

$$T=523.15K$$

$$P=7000kPa$$

$$H=1085.03 \text{ KJ/kg}$$

$$S=2.7844 \text{ kJ/kgK}$$

We can solve for G_2

$$G_2 = 1085.03 \frac{\text{kJ}}{\text{kg}} - (523.15K) \left(2.7844 \frac{\text{kJ}}{\text{kg} \cdot K} \right) = -371.629 \frac{\text{kJ}}{\text{kg}}$$

We convert to J/mol by using the molar mass of water (18.0152 g/mol)

$$G_2 = -371.629 \frac{\text{kJ}}{\text{kg}} * \frac{1000J}{1kJ} * \frac{1kg}{1000g} * \frac{18.0152g \text{ of water}}{mol} = -6694.97 \frac{J}{mol}$$

$$G_2 = -6694.97 \frac{J}{mol}$$

For a low pressure we can use 10kPa and temperature 250°C and using the steam tables we find

$$T=523.15K$$

$$P=10kPa$$

$$H=2975.45 \text{ kJ/kg}$$

$$S=9.0949 \text{ kJ/kg}\cdot K$$

We can solve for G_1

$$G_1 = 2975.45 \frac{kJ}{kg} - (523.15K) \left(9.0949 \frac{kJ}{kg \cdot K} \right) = -1782.55 \frac{kJ}{kg}$$

We convert to J/mol

$$G_1 = -1782.55 \frac{kJ}{kg} * \frac{1000J}{1kJ} * \frac{1kg}{1000g} * \frac{18.0152g \text{ of water}}{mol} = -32112.9 \frac{J}{mol}$$

$$G_1 = -32112.9 \frac{J}{mol}$$

We can now use eq.4 to solve for the fugacity (f_2) by using $8.314 \text{ J/mol}\cdot K$

We can substitute f_1 with $10kPa$ since $f=P$ as pressure approaches 0

$$G_2 - G_1 = RT \ln \left(\frac{f_2}{f_1} \right) \text{ (eq. 4)}$$

$$-371.629 \frac{kJ}{kg} - \left(-32112.9 \frac{J}{mol} \right) = \frac{8.314J}{mol \cdot K} * 523.15K * \ln \left(\frac{f_2}{10kPa} \right)$$

$$f_2=3451.31kPa$$

We can convert to atm ($1kPa=0.00987 \text{ atm}$)

$$f_2=34.51 \text{ atm}$$

The fugacity for water at $250^\circ C$ and 70 bar is
 $f_2=34.51 \text{ atm}$

- 2) We can solve for the fugacity assuming that water follows the Peng-Robinson EOS in both gas and liquid phase by using eq.3 from the previous problem.

$$G - G^{IG} = RT \ln \frac{f}{P} \quad (eq. 3)$$

$\frac{f}{P}$ becomes the fugacity coefficient denoted by Φ

So now we have

$$\Phi = \frac{f}{P} \quad (eq. 5)$$

Water is in both gas and liquid phase so we have to solve for the fugacities and both the liquid and gas phase. We can do that by obtaining the Φ values for the gas and liquid phases from the CHE 301 calculator functions and solving for f .

1. We first set the Calculator to Peng-Robinson EOS.
2. We then add our component which is water.
3. We can input the temperature as 250° and pressure as 70 bar
4. We first solve for the compressibility factor using the zcomp function

For the gas phase we input "=ZComp(comp1,Temp,Pres,1)"

$$Z=0.69882$$

For the liquid phase we input "=ZComp(comp1,Temp,Pres,0)"

$$Z=0.0448$$

5. We can now solve for the fugacity coefficient using the PhiC function

For the gas phase we input "=PhiC(comp1,comp1,Temp,Pres,C19)"

C19 is the compressibility factor for the gas phase

$$\Phi = 0.77156$$

For the liquid phase we input "=PhiC(comp1,comp1,Temp,Pres,C30)"

C30 is the compressibility factor for the liquid phase

$$\Phi = 0.50763$$

6. We can now solve for the fugacity values for both liquid and gas by rearranging eq.5

into

$$f = \Phi * P$$

For the gas phase

$$f=0.77156 \cdot 50 \text{ bar}$$

$$f=54.00944 \text{ bar}$$

For the liquid phase

$$f=0.50763 \cdot 50 \text{ bar}$$

$$f=35.35582 \text{ bar}$$

We now get 2 different fugacity values but the fugacity when water is both in the liquid and gas phase will be the lower fugacity value because when we have a lower fugacity we have more stability.

Therefore,

$$f= 35.53382 \text{ bar}$$

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T24

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33

VLE Calculations

Peng-Robinson

Fluid Conditions	Mixture	H2O																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
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3) To solve for the fugacity of water that follows the Peng Robinson EOS in only the gas phase we can use the Poynting Formula because

$$\Phi^l = \Phi^{sat} * \frac{P^{sat}}{P} * \exp \left[\frac{V^l}{RT} (P - P^{sat}) \right]$$

And we know that

$$\Phi(l)_{sat} = \Phi(v)_{sat}$$

So we can say

$$\Phi^v = \Phi^{sat} * \frac{P^{sat}}{P} * \exp \left[\frac{V^v}{RT} (P - P^{sat}) \right]$$

1. We start by solving for Psat first. We can use the EOSPsat function for the Psat and we will use the Psat function in the calculator for the initial pressure guess for the EOSPsat (all pressures were converted to atm [1 bar=0.987 atm])

$$Psat = Psat(comp1, T4) * 0.987$$

$$Psat = 36.5965 \text{ atm}$$

$$EOSPsat = EOSPsat(comp1, T4, T5) * 0.987$$

$$EOSPsat = 39.5977 \text{ atm}$$

2. In order to solve for fugacity coefficient of the gas we first must solve for the compressibility factor of the gas. We can solve for the compressibility factor (sat) by using the Zcomp function in the calculator

$$= ZComp(comp1, T4, T6, 1) \text{ (we use the EOSPsat pressure for the pressure in the function)}$$

$$Zcomp(v) = 0.8529$$

3. We can now solve for the fugacity coefficient (sat) by using the PhiC function in the calculator

$$PhiCsat = PhiC(comp1, comp1, T4, T6, T8)$$

$$PhiCsat = 0.8705$$

where we use EOSPsat for the pressure and the Z value for the gas phase

4. We now find the volume "V(v)" using the Rackett equation from the calculator

$$V = 1 / Rackett$$

$$V = 1 / Rackett(comp1, T4)$$

$$V = 0.0223$$

5. We have all the components that we need to solve the Poynting Formula ($R = 0.08206 \text{ L} \cdot \text{atm} / \text{mol} \cdot \text{K}$ and $T = 523.50 \text{ K}$). We input everything into excel.

$$\Phi^v = 0.8705 * \frac{39.5977 \text{ atm}}{69.0900 \text{ atm}} * \exp \left[\frac{0.0223 \text{ L}}{(0.08206 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}})} (69.0900 \text{ atm} - 39.5977 \text{ atm}) \right]$$

$$\Phi^v = 35.0035 \text{ atm}$$

The fugacity for water in the gas phase is

$$\Phi = 35.0035 \text{ atm}$$

P=	✓	69.0900 atm
T=	✓	250.0000 °C
Psat=	✓	36.5965 atm
EOspsat=	✓	39.5977 atm
zcomp(l)=	✓	0.0255
zcomp(v)=	✓	0.8520
phiC sat=	✓	0.8705
V(l)=	✓	0.0223 L
Phi(l)	✓	35.0035 atm
Psat/P	✓	0.5731
V(l)/RT	✓	0.0005
P-Psat	✓	29.4923 atm
R=		0.0821 L*atm/mol*K

Problem 5B)

- 1) To find whether Propane that follows the van der Waals EOS is in the liquid or gas phase or both at once we must determine if propane lies below or above the saturated pressure with our given temperature and pressure.

1. We start by using the Psat function using our given temperature in the 301 calculator to find the saturated pressure of Propane, we will use this as our initial guess for when we use the EOSPsat function

$$=Psat(comp1,C4)$$

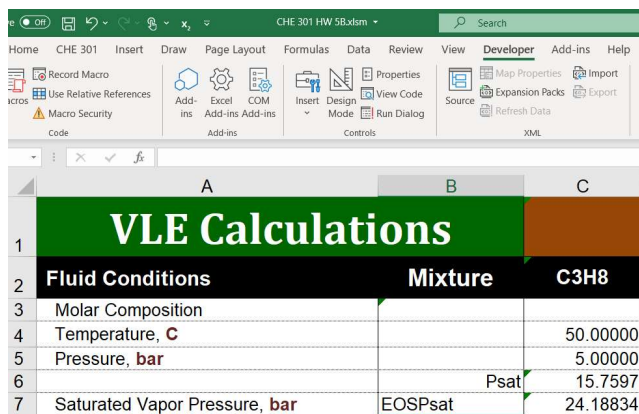
$$Psat=15.7597 \text{ bar}$$

2. Since Propane follows an equation of state, we must use the EOSPsat to find the saturated pressure of Propane that follows the van der Waals EOS.

$$=EOSPsat(comp1,C4,C6)$$

$$EOSPsat=24.18834 \text{ bar}$$

Since the saturated pressure of Propane that follows the van der Waals EOS is above the saturated pressure of Propane that doesn't follow an EOS we can say that Propane that follows van der Waals EOS at 50°C and 5 bar is in the gas phase.



	A	B	C
1	VLE Calculations		
2	Fluid Conditions	Mixture	C3H8
3	Molar Composition		
4	Temperature, C		50.00000
5	Pressure, bar		5.00000
6		Psat	15.7597
7	Saturated Vapor Pressure, bar	EOSPsat	24.18834

2) To calculate the value of the derivative $\left. \frac{\partial T}{\partial P} \right|_S$ we must first find an equation that satisfies the derivative.

1. We can start with

$$dH = TdS + VdP$$

2. We rearrange that equation and we get

$$TdS = dH - VdP$$

3. Since $dH = C_p dT$ we can substitute the equation with

$$TdS = C_p dT + (1 - \beta T)VdP - VdP$$

4. When we rearrange we get

$$TdS = C_p dT - \beta TVdP$$

5. Divide by T and we get

$$dS = \frac{C_p}{T} dT - \beta VdP \quad (\text{eq.1})$$

6. Propane is an ideal gas so we can say that $\beta = 1/T$ and $V = R/P$ and we can substitute this into eq.1 and we get

$$dS = \frac{C_p}{T} dT - \frac{R}{P} dP$$

7. We can divide by dP and we get

$$dS/dP = \frac{C_p}{T} \frac{dT}{dP} - \frac{R}{P}$$

8. We are holding entropy constant so we get

$$0 = \frac{C_p}{T} \left. \frac{\partial T}{\partial P} \right|_S - \frac{R}{P}$$

9. We rearrange the equation and we get

$$\left. \frac{\partial T}{\partial P} \right|_S = \frac{RT}{PC_p}$$

10. For the as constant we will use $R = 8.314 \text{ J/mol} \cdot \text{K}$

11. We can plug this into the 301 excel all together

$$= (8.314 * T_9) / (T_4 * (\text{CPIG}(\text{comp1}, 50)))$$

Where pressure was converted to atm and temperature into Kelvin and C_p is replaced with the calculator function CpIG . The equation with units is:

$$\frac{\left(8.314 \frac{\text{J}}{\text{molK}}\right) (323.15\text{K})}{(4.9350\text{atm}) \left(C_p^{IG} \frac{\text{kJ}}{\text{kmolK}}\right)} \cdot \frac{1\text{kJ}}{1000\text{J}} \cdot \frac{1000\text{mol}}{1\text{kmol}} * \frac{1\text{atm}}{1.01325\text{bar}}$$

12. After cancelling units, the final answer is:

$$\left. \frac{\partial T}{\partial P} \right|_s = 6.77855 \text{ K/bar}$$

3) To calculate the numerical value of the derivative $\left. \frac{\partial T}{\partial P} \right|_S$ by using first order numerical differencing for Propane that follows the van der Waals EOS we start by setting the CHE 301 calculator to van der Waals

1. We can now add Propane as a component and input the temperature as 50°C and the pressure as 5 bar.

2. We can now find the compressibility factor of the vapor using the Zcomp function in the calculator using the given temperature and pressure and setting Z(0) to 1 for vapor

$$=ZComp(comp1,C4,C5,1)$$

$$Z=.94954$$

3. Since we hold entropy constant and we are using first order differencing we can set the difference between 2 entropies to 0. We do that by estimating another value for pressure. I chose to increase the pressure by .001 so we can say P2=5.0100 bar

4. We can set the temperature (T2) for P2 as 51°C, we will change that later.

5. We now calculate Zcomp using the function in excel and we get

$$Z=.94982$$

6. We can now find 2 different entropies using the entropy function in excel

$$=Entropy(comp1,T,P,Z)$$

$$S=-277.44705 \text{ at } T=50^\circ\text{C and } P=50 \text{ bar}$$

$$S=-277.21576 \text{ at } T=51^\circ\text{C and } P=5.01 \text{ bar}$$

7. There is a slight difference between the 2 entropies but since we are holding entropy constant we have to make sure the two entropies are the same. We can set the two entropies equal to each other by using the extended secant method

8. We use the extended secant method and make the difference between the cells holding the entropy values 0 by changing T2. The value for temperature (T2) changes to 50.07165 °C. We can not find the numerical value for $\left. \frac{\partial T}{\partial P} \right|_S$.

9. The formula for first order numerical differencing is

$$\frac{df}{dx} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

We can substitute that formula to satisfy our derivative and we get

$$\left. \frac{\partial T}{\partial P} \right|_S = \frac{T(P+\Delta P) - T(P)}{\Delta P}$$

10. Now we can use the values for temperature and pressure that we have into that formula

$$\left. \frac{\partial T}{\partial P} \right|_S = \frac{50.07165 - 50}{5.01 - 5}$$

And we get:

$$\left. \frac{\partial T}{\partial P} \right|_S = 7.16532 \text{ K/bar}$$

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Main Numerical Methods Vapor Fraction Calculations Bubble Point Calculations Dew Point Calculations Excel Commands AthenaVisual Studio

Temp

	A	B	C	D	E
1	VLE Calculations				
2	Fluid Conditions	Mixture	C3H8		
3	Molar Composition				
4	Temperature, C		50.00000	50.07165	
5	Pressure, bar		5.00000	5.01000	
6					
7	Saturated Vapor Pressure, bar				
8	Saturated Vapor Temperature, C				
9	density				
10	K-Values		dT/dP S=	7.16532	
11	New K-Values				
12					
13	Vapor to Feed Molar Ratio (a=V/F)				
14	The Rachford-Rice Equation				
15	Discriminant of the Cubic EOS				
16	Vapor Phase				
17	Molar Composition				
18	New Molar Composition				
19	Compressibility Factor		0.94954	0.94946	
20	Fugacity Coefficient		0.95189		
21	Saturated Fugacity Coefficient				
22					
23	Molar Enthalpy, kJ/kmol				
24	Molar Entropy, kJ/(kmol K)		-277.44705	-277.44705	
25	Molar Density, kmol/m ³				

Data Base VLE Calculations ENG Cal ...

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4) To develop an expression for $\left. \frac{\partial T}{\partial P} \right|_S$ for the van der Waals EOS we can start with

$$dH = TdS + VdP$$

1. We rearrange that equation and we get

$$TdS = dH - VdP$$

2. Since $dH = C_p dT$ we can substitute the equation with

$$TdS = C_p dT + (1 - \beta T)VdP - VdP$$

3. When we rearrange we get

$$TdS = C_p dT - \beta TVdP$$

4. Divide by T and we get

$$dS = \frac{C_p}{T} dT - \beta VdP$$

5. We can divide by dP and we get

$$\frac{dS}{dP} = \frac{C_p}{T} \frac{dT}{dP} - \beta V$$

6. We are holding entropy constant so the equation is now

$$0 = \frac{C_p}{T} \frac{dT}{dP} - \beta V$$

7. And we can rearrange

$$\left. \frac{\partial T}{\partial P} \right|_S = \frac{\beta VT}{C_p}$$

8. To get a numerical value for the derivative we can use the CHE 301 calculator by setting it to van der Waals EOS and using properties of Propane

9. After adding Propane as a component we can find the compressibility factor

$$Z = Z_{\text{comp}}(\text{comp1}, T, P1)$$

$$Z = .94954$$

10. We find volume by using the density formula in excel

- a. $\rho = \text{density}(T, P, Z)$ where T is in °C and P in bar

$$\rho = .19598 \text{ kmol/m}^3$$

- b. Now volume will be $V = 1/\rho$

$$V = 5.10250 \text{ m}^3/\text{kmol}$$

11. We can find β using the Zprop function in the calculator

$$\beta = Z_{\text{prop}}(5, \text{comp1}, T, P, Z) \text{ where T is in } ^\circ\text{C} \text{ and P in bar}$$

$$\beta = 0.00350 \text{ 1/K}$$

12. We can find C_p using the Zprop function in the calculator

$$C_p = (2, \text{comp1}, T, P, Z)/100 \text{ where T is in } ^\circ\text{C} \text{ and P in bar}$$

$$C_p = 80.530 \text{ kJ/(kmol}\cdot\text{K)}$$

13. We now have everything that we need to solve for our equation

$$\left. \frac{\partial T}{\partial P} \right|_s = \frac{(0.00350 K^{-1}) \left(5.10250 \frac{m^3}{kmol} \right) (323.15 K)}{\left(80.53046 \frac{kJ}{kmol \cdot K} \right)} = 0.07172 \frac{K \cdot m^3}{kJ}$$

14. We have to cancel some units

$$0.07172 \frac{K \cdot m^3}{kJ} * \frac{1 kJ}{1000 J} * \frac{1000 J}{1000 N \cdot m} \frac{1000 N \cdot m}{1 kPa \cdot m^3} * \frac{1 kPa}{.01 bar} = 7.17203 \frac{K}{bar}$$

Our final answer is:

$$\left. \frac{\partial T}{\partial P} \right|_s = 7.17203 \frac{K}{bar}$$

Van Der Waals		
Density=	0.19598	kmol/m³
Volume=	5.10250	m³/kmol
Cp	0.00350	kJ/(kmol·K)
Beta	80.53046	1/K
dT/dP S	7.17203	K/bar

5)

Method	$\left. \frac{\partial T}{\partial P} \right _s \frac{K}{bar}$
Ideal Gas	$\left. \frac{\partial T}{\partial P} \right _s = 6.77855 \text{ K/bar}$
first order numerical differencing (assuming Propane follows van der Waals)	$\left. \frac{\partial T}{\partial P} \right _s = 7.16532 \text{ K/bar}$
Van der Waals EOS	$\left. \frac{\partial T}{\partial P} \right _s = 7.17203 \frac{K}{bar}$